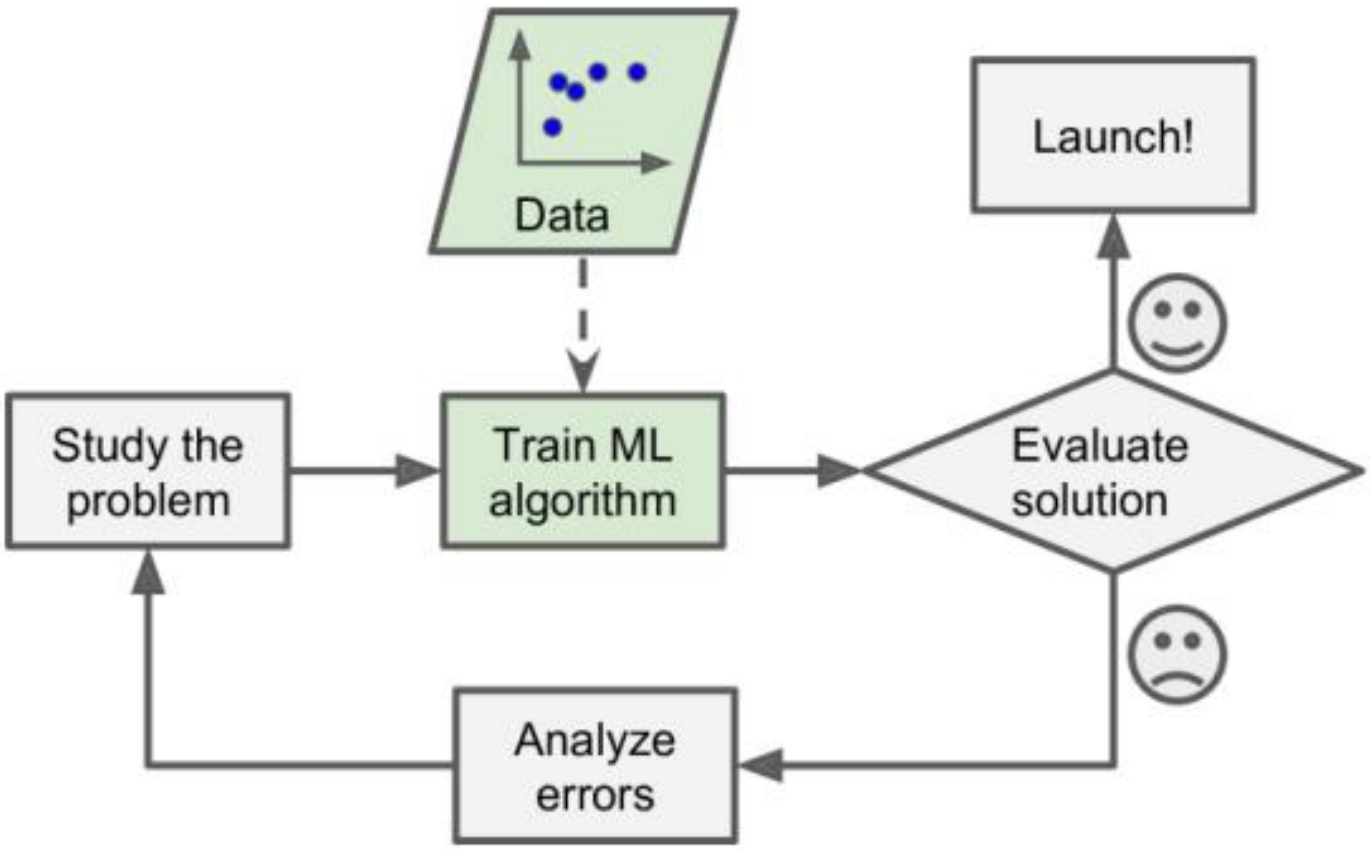
Lecture 1
Introduction to Machine Learning

- What is machine learning?
- History and relationships to other fields
- Applications of machine learning
- Types of machine learning techniques
- Overview of Data mining and KDD process:
- Prediction vs. Description modeling

Definition of Machine Learning (ML)

- ML is a field of computer science with the help of which computer systems can provide sense to data in much the same way as human beings do.
- It extracts patterns out of raw data by using an algorithm or method.
- It learns from experience without being explicitly programmed or human intervention.

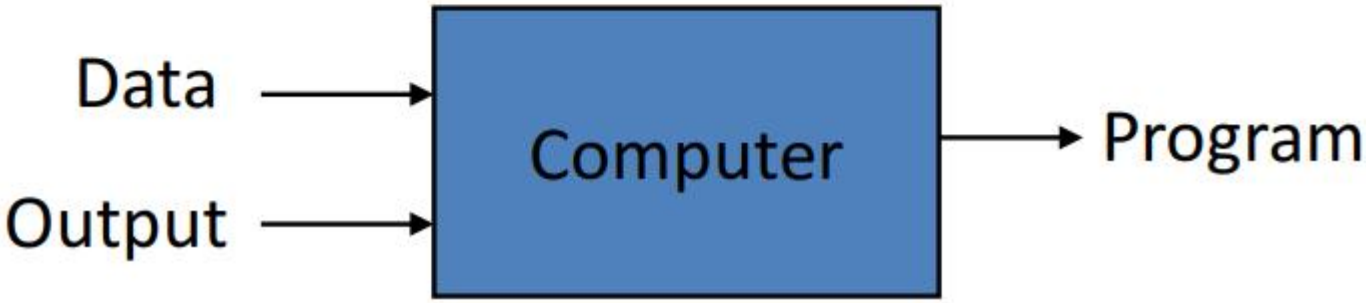
Machine Learning Approach



Traditional Programming vs. Machine Learning

Traditional Programming

Machine Learning



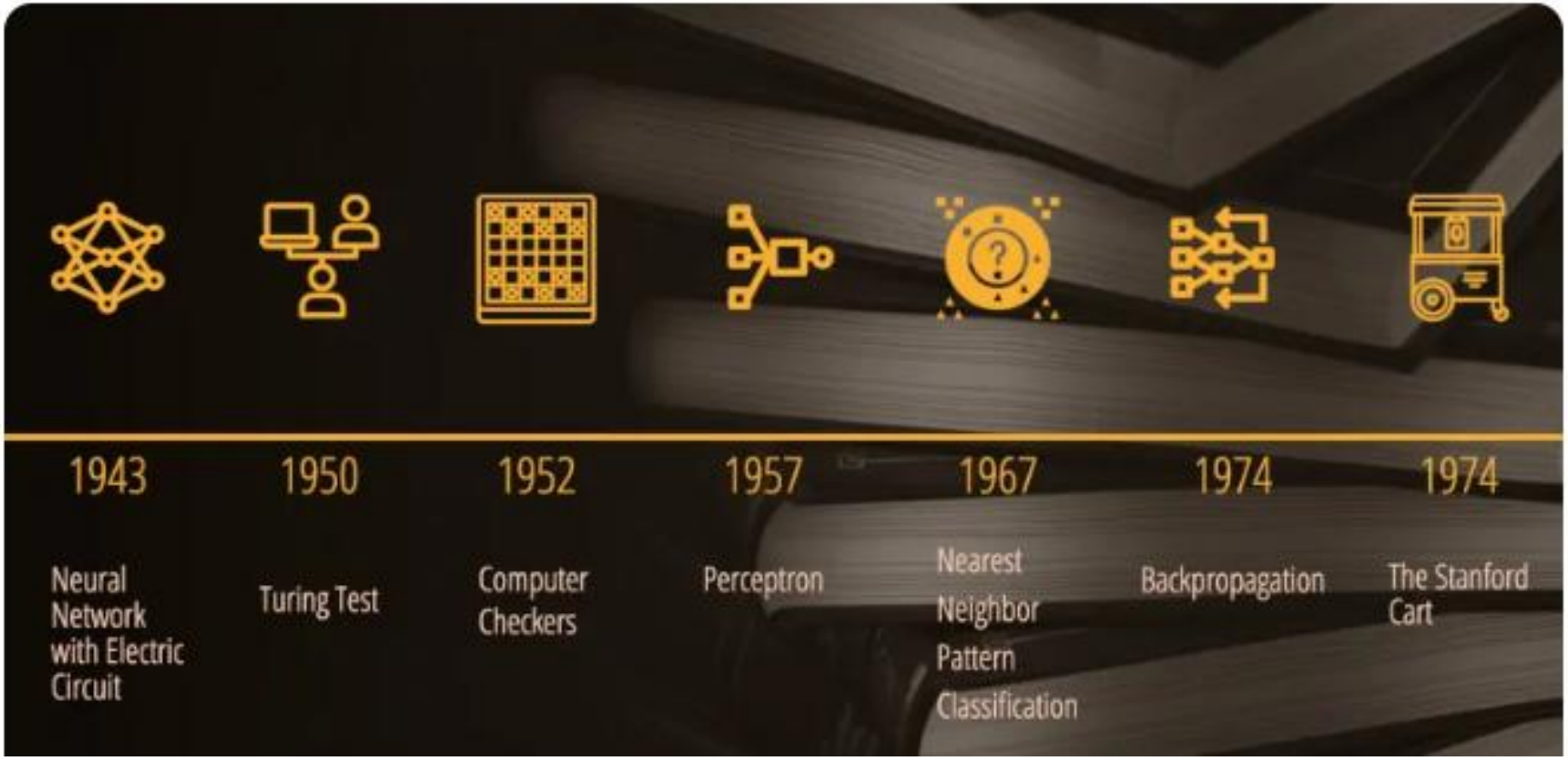
History of Machine Learning

- Machine learning grew out of the quest for artificial intelligence.
- Already in the early days of AI as an academic discipline, some re-searchers were interested in having machines learn from data then termed "neural networks“.
- However, an increasing emphasis on the logical, knowledge-based approach caused a rift between AI and machine learning.

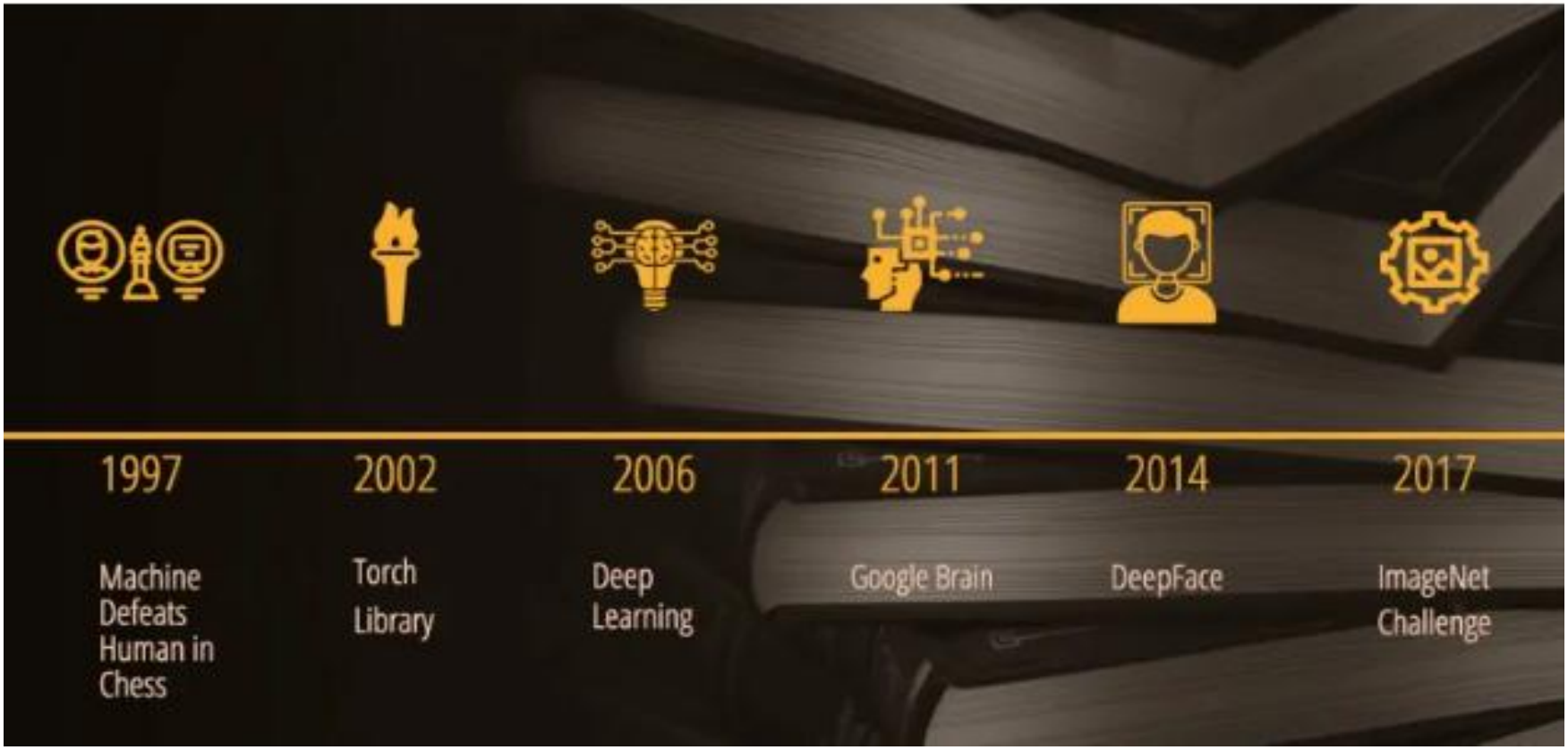
History of Machine Learning

- Machine learning, reorganized as a separate field, started to flourish in the 1990s.
- The field changed its goal from achieving artificial intelligence to tackling solvable problems of a practical nature.
- It shifted focus away from the symbolic approaches toward methods and models borrowed from statistics and probability theory.

Early History of Machine Learning



History of Machine Learning



Present: State-of-the-art Machine Learning

- Machine learning is used in many different fields, from fashion to agriculture.
- Machine Learning algorithms are able to learn patterns and relationships between data, find predictive insights for complex problems and extract information that is otherwise too difficult to find.
- Today's Machine Learning algorithms are able to handle large amounts of data with accuracy in a relatively short amount of time.

Machine Learning and Data Mining

- Both employ the same methods and overlap significantly.
- They can be roughly distinguished as follows:
  - ✓ Machine learning focuses on prediction, based on known properties learned from data.
  - ✓ Data mining focuses on the discovery of (previously) unknown properties in the data.
- Machine learning also has intimate ties to optimization:

Types of Machine Learning

- Classification of ML in broad categories based on:
  - ✓ Whether or not they are trained with human supervision (supervised, unsupervised, semi-supervised, and Reinforcement Learning)
  - ✓ Whether or not they can learn incrementally on the fly (online versus batch learning)

Types of Machine Learning

1 Supervised Learning

- ✓ Deals with labeled datasets
- ✓ Tow processes
  - ✓ Classification
  - ✓ Regression

2 Un-supervised Learning

- ✓ It uses unlabeled data
- ✓ No idea which types of results are expected.
- ✓ Two types
  - ✓ Clustering
  - ✓ Dimensionality reduction

3 Semi-supervised Learning

- ✓ Known as hybrid learning and it lies between supervised and unsupervised learning.
- ✓ Combination of labeled and unlabeled data.

4 Reinforcement Learning

- ✓ There are no training data sets.
- ✓ An example of reinforcement learning problem is playing game.

Supervised Machine Learning

- In supervised learning the machine experiences the examples along with the labels or targets for each example. The labels in the data help the algorithm to correlate the features.
- It used labeled dataset (Dataset with Class label)
- Example classification and Prediction (regression)
  - ✓ Regression: output value is continuous.
  - ✓ Classification: output value is discrete or "categorical"

Unsupervised Machine Learning

- The dataset used is not “labeled”
- Extract the pattern (cluster) from the available data.
- Example Clustering and Association Rule Discovery
  - ✓ **Clustering**: finding “groups” in the data, according to a similarity measure
  - ✓ **Association Rule Mning**: Discover rules that describe large portions of your data, such as people who buy X also tend to buy Y.

Semi-Supervised Learning

- Hybrid learning architectures combine supervised and unsupervised learning components.
- In semi-supervised learning, training data is split into two. A small amount of labeled data and a larger set of unlabeled data. In this case, the model uses labeled data as an input to make inferences about the unlabeled data, providing more accurate results than regular supervised-learning models.
- This approach is gaining popularity, especially for tasks involving large datasets such as image

Reinforcement Learning

- Reinforcement learning (RL) is concerned with how a software agent (or computer program) ought to act in a situation to maximize the reward.
- RL attempts to determine the best possible path it should take in a given situation.
- This machine learning method is mostly used in robotics and gaming. Video games demonstrate a
- clear relationship between actions and results, and can measure success by keeping score.

The Machine Learning Process

The Machine Learning Process

- Problem Framing

- Decide what kind of problem are you trying to solve.
  - ✓ Develop a model to classify emails as spam or not spam
  - ✓ Develop a model to classify tumor as malignant or benign,
  - ✓ Develop a model to classify English language to Amharic.
  - ✓ Pattern discovery that shows the association of items in supermarket shelves.

The Machine Learning Process

- Data Analysis
- Collect the dataset available for developing the model.
  - ✓ Data visualization of features,
  - ✓ Handling missing data, handling categorical data,
  - ✓ Encoding class labels,
  - ✓ Normalization, and standardization of features,
  - ✓ Feature engineering,
  - ✓ Dimensional reduction,
  - ✓ Data partitioning into training, validation and testing sets, etc.

The Machine Learning Process

- Model Building
- This is the core part of machine learning by selecting appropriate ML algorithms.
  - ✓ linear regression, logistic regression, KNN, SVM, K-means, apriori algorithm, time series analysis, etc.
  - ✓ The model developed from the training dataset
  - ✓ Performance is evaluated using the testing dataset
- Performance Evaluation
  - ✓ Accuracy, Precision, Recall
  - ✓ Support, Confidence

The Machine Learning Process

- Application
- The final machine learning model is put into production to assist decision-making in
  - ✓ Increasing productivity
  - ✓ Deciding if a bank should approve credit to a borrower, etc.

Challenges in Machines Learning

- The major challenges that ML is facing currently are:
  - ✓ No clear objective for formulating business problems
  - ✓ Quality of dataset.
  - ✓ Time-Consuming task:
  - ✓ Lack of specialist persons:.
  - ✓ Issue of overfitting & underfitting
  - ✓ Curse of dimensionality
  - ✓ Difficulty in deployment

Applications of Machines Learning

- Sentiment analysis
- Image Recognition
- Virtual Assistants
- Product Recommendations
- Medical Diagnosis
- Network intrusion detection
- Weather forecasting and prediction
- Stock market analysis and forecasting
- Speech recognition
- Customer segmentation
- Object recognition
- Fraud detection

Assignment-1

- Take a research paper (thesis) of a machine learning application and write the following
  - ✓ Title of the research
  - ✓ Objective of the research
  - ✓ From where the dataset collected
  - ✓ What preprocessing techniques applied on the dataset
  - ✓ What a machine learning technique(s) applied on it
  - ✓ What a performance result has been obtaine

Revision of Basic Statistics

- **Reading Assignment**
  - Mean, Median, Mode
  - Max, Min, Range
  - Variance, Standard Deviation
  - Quartiles, Interquartile range
  - Histogram, Scatter plot
  - Symmetric data, Positively skewed, Negatively skewed

Python Library (Pandas)

- It is one of the most famous Python libraries and used for importing and managing the datasets.
- It is an open-source data manipulation and analysis library.
- It will be imported as below:

```
import pandas as pd
```

Python Library (Numpy)

- Numpy Python library is used for including any type of mathematical operation in the code.
- It is the fundamental package for scientific calculation in Python.
- It also supports to add large, multidimensional arrays and matrices. So, in Python,
- Imported as:

```
import numpy as np
```

Python Library (Matplolib)

- It is a Python 2D plotting library.
- We need to import a sub -library pyplot.
- This library is used to plot any type of charts in Python for the code. It will be
- imported as below:
- import matplotlib.pyplot as plt

Python Library (Pandas)

- Python toolboxes/libraries for data processing:
  - ✓ NumPy
  - ✓ SciPy
  - ✓ Pandas
- Visualization libraries
  - ✓ matplotlib
  - ✓ Seaborn
- Machine learning & deep learning
  - ✓ Scikit-learn ○
  - ✓ Tensorflow/Pytorch/Theano